SlimDy-YOLO: A Lightweight Dynamic Detector for Pediatric Wrist Fracture Detection in X-ray Images

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Abstract. Detecting pediatric wrist fractures in X-ray images requires localizing weak radiographic evidence under real-time screening constraints. Existing high-capacity detectors improve representation but are costly, while lightweight YOLO-style detectors may miss subtle fracture cues or confuse them with growth-plate-like structures when feature extraction and fusion remain generic. We focus on two problems: adaptive fracture-evidence extraction and efficient multi-scale evidence preservation. SlimDy-YOLO addresses these problems by combining D-HGNet for input-adaptive backbone representation, Slim-Neck for low-cost multi-scale fusion, and DySample for content-aware upsampling of small fracture-related cues. On the GRAZPEDWRI-DX dataset, SlimDy-YOLO achieves 68.47% mAP_{50} with 40.5 GFLOPs and 16.83M parameters, while reaching 265.96 FPS under the tested inference setting. These results suggest that coordinated backbone, neck, and upsampling design can provide an efficient task-specific detector for pediatric wrist fracture screening.

Keywords: Pediatric wrist fracture · X-ray detection · YOLOv11 · Lightweight detector

1 Introduction

Pediatric wrist fracture detection is a fine-grained radiographic detection problem under clinical time constraints. Wrist fractures are common in pediatric emergency care, and missed or delayed diagnosis may lead to malunion or growth-related complications [7]. X-ray radiography is therefore widely used for first-line screening, but pediatric wrist images are difficult to interpret: growth plates may resemble fracture lines, subtle greenstick or avulsion fractures can have low contrast, and image appearance varies across acquisition conditions [9, 2]. In this setting, a useful detector must localize weak fracture evidence, reject visually similar anatomical structures, and remain fast enough for emergency or resource-limited screening [15].

Existing studies have addressed this requirement from several directions. High-capacity detectors improve representation and localization: for example, Zech et al. used Faster R-CNN for pediatric wrist fracture detection, while Yan et

al. introduced a wavelet-enhanced hierarchical DETR-style detector for stronger end-to-end modeling [25, 23]. One-stage detectors improve practical efficiency: Ahmed et al. compared YOLOv5–YOLOv8 with Faster R-CNN on wrist abnormality detection, and Liu and Zhang further evaluated one-stage and two-stage frameworks for fracture detection in X-ray images [1, 13]. More recent YOLO-based variants modify attention, convolution, or fusion modules, including YOLOv8-AM, feature-context excitation YOLOv8, G-YOLOv11, FracDet-v11, and SE-enhanced YOLO11 [4, 8, 5, 16, 18]. These studies show that pediatric wrist fracture detection has moved from generic detection frameworks toward task-specific YOLO adaptations.

However, these representative directions still leave a mechanism-level gap. High-capacity detectors improve modeling ability but are less attractive for real-time screening. YOLO-style detectors are efficient, but their generic fixed-convolution backbones and standard feature pyramids are not specifically designed for weak radiographic evidence. Existing YOLO variants often strengthen one component, such as attention, lightweight convolution, wavelet processing, or feature fusion, yet they rarely treat the backbone, neck, and upsampling path as a coordinated response to two needs of pediatric wrist radiographs: adapting to image-dependent fracture appearance and preserving small low-contrast cues across scales. Thus, the remaining difficulty is not fully captured by a generic trade-off between accuracy and efficiency.

We therefore formulate two core problems for lightweight pediatric wrist fracture detection. The first is *adaptive fracture-evidence extraction*: how to make the backbone respond to radiograph-specific fracture patterns without introducing a heavyweight global module. The second is *efficient multi-scale evidence preservation*: how to fuse and upsample shallow structural cues and deeper semantic context while keeping computation suitable for real-time screening. Solving these two problems provides a more targeted route to this task than treating the model as a generic lightweight detector.

To address these problems, we propose SlimDy-YOLO, a YOLOv11-based detector for pediatric wrist X-ray images. D-HGNet uses dynamic convolution within a PPHGNetV2-inspired backbone [3, 24], allowing feature extraction to adapt to input radiographs. Slim-Neck uses GSConv and VoVGSCSP to reduce redundant computation during multi-scale fusion while maintaining interaction between shallow and deep features [10]. DySample replaces fixed interpolation with content-aware sampling so that upsampled feature maps better preserve small fracture-related cues [12]. A radiograph-oriented augmentation strategy is further used to improve robustness to contrast, brightness, noise, and blur variations.

The main contributions are summarized as follows:

(1) We recast pediatric wrist fracture detection as weak radiographic evidence detection under real-time constraints, emphasizing adaptive evidence extraction and efficient multi-scale evidence preservation.

(2) We propose SlimDy-YOLO, a lightweight YOLOv11-based framework that aligns the backbone, neck, and upsampling path with the visual characteristics of pediatric wrist radiographs.
(3) We introduce D-HGNet for adaptive fracture-evidence extraction and combine Slim-Neck with DySample for efficient multi-scale evidence preservation.
(4) Experiments on GRAZPEDWRI-DX show that SlimDy-YOLO improves mAP_{50} over the YOLOv11m baseline while reducing GFLOPs and parameters, providing a favorable balance between detection accuracy and efficiency.

2 Related Work

Public pediatric wrist datasets have made fracture detection a concrete object detection problem. GRAZPEDWRI-DX provides multi-class annotations for pediatric wrist trauma radiographs and has become a common benchmark for detector comparison [14]. Early work used high-capacity detectors to improve localization, such as Faster R-CNN for pediatric wrist fractures [25]. More recent work has also explored end-to-end or Transformer-style designs, including wavelet-enhanced hierarchical matching for wrist fracture detection [23]. These methods improve representation and region modeling, but their cost can be less suitable for real-time screening. In contrast, YOLO-style detectors provide practical one-stage baselines. Ahmed et al. compared YOLOv5–YOLOv8 with Faster R-CNN for wrist abnormality detection, and Liu and Zhang evaluated multiple one-stage and two-stage detectors for X-ray fracture detection [1, 13].

YOLO-based adaptations have therefore become a major direction for pediatric wrist fracture detection. Attention-enhanced designs improve discriminative feature learning: Chien et al. embedded a residual CBAM design into YOLOv8l, and Ju et al. introduced feature-context excitation modules for YOLOv8 [4, 8]. YOLOv11 variants further explore the cost-accuracy balance. Ferdi used GhostConv and C3Ghost to reduce computation in G-YOLOv11, while Qiu et al. proposed FracDet-v11 with wavelet, attention, and deformable-convolution components [5, 16]. Tariq and Choi used an SE_BLOCK-enhanced YOLO11 design, and Zhang and Ji introduced a Bone-MSCA module with a modified regression loss [18, 26]. These methods show that task-specific modifications can improve pediatric wrist fracture detection, but they often emphasize either stronger attention and capacity or more aggressive compression.

Feature fusion and upsampling are also important for weak radiographic evidence because fracture cues may appear as small, low-contrast structures across scales. General multi-scale fusion designs, such as BiFPN, MAFPN, GFPN, and spatial feature calibration modules, improve feature aggregation but may add cost or be designed for broader detection and segmentation settings [17, 27, 22, 11]. Lightweight neck designs such as Slim-Neck use GSConv to reduce redundant computation in real-time detectors [10]. At the upsampling stage, content-aware operators such as CARAFE and sampling-based approaches such as DySample show that learned or content-dependent resampling can better reconstruct

high-resolution features than fixed interpolation in some vision tasks [19, 12]. These studies motivate treating fusion and upsampling as part of the evidence-preservation problem rather than as incidental implementation choices.

In summary, prior work improves pediatric wrist fracture detectors from individual directions: stronger high-capacity modeling, faster YOLO-style inference, attention-based enhancement, lightweight convolution, and content-aware feature processing. Few designs, however, coordinate the backbone, feature-fusion path, and upsampling operator around the two requirements of adaptive fracture-evidence extraction and efficient multi-scale evidence preservation. SlimDy-YOLO is designed to address this gap with a dynamic backbone, a lightweight neck, and content-aware upsampling under real-time screening constraints.

3 Proposed Method

3.1 Overall Architecture of SlimDy-YOLO

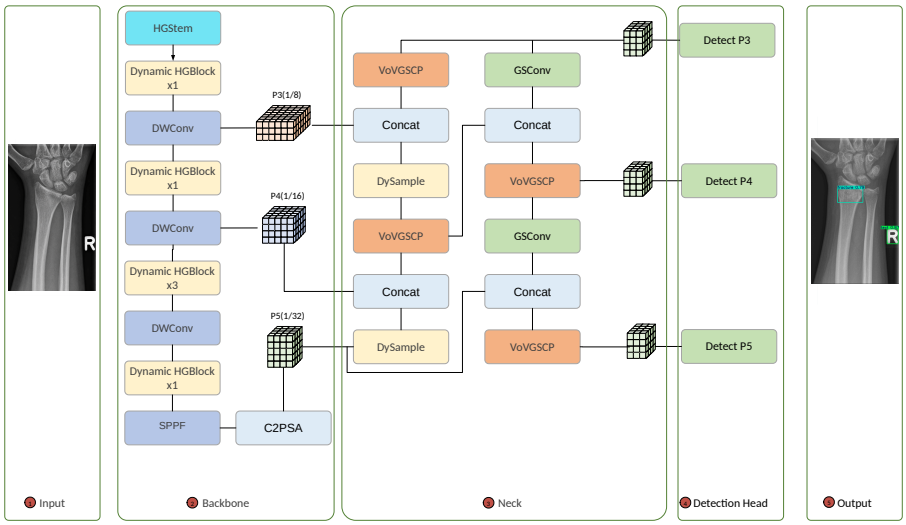


Fig. 1: Overall architecture of SlimDy-YOLO.

As shown in Fig. 1, SlimDy-YOLO keeps the YOLOv11 detection head and redesigns the feature extraction and fusion path around the two problems defined in the Introduction. For adaptive fracture-evidence extraction, we replace the Conv/C3k2-based backbone with D-HGNet, a PPHGNetV2-inspired backbone [3] that uses Dynamic_HGBlocks [24] to condition convolutional responses on each radiograph. This input-adaptive design aims to make the backbone more responsive to low-contrast fracture patterns and visually similar anatomical structures without adding a heavyweight attention module. For efficient multi-scale evidence preservation, we use Slim-Neck [10] and DySample [12] in the

neck. GSConv and VoVGSCSP reduce dense full-channel computation through grouped convolution, channel shuffling, and CSP-style partial-channel aggregation, while DySample replaces fixed interpolation with content-aware sampling offsets for high-resolution feature alignment. The detection head then produces multi-scale predictions at P3/P4/P5.

Given an input radiograph $\mathbf{X}$, the dynamic backbone $\mathbf{B}_{dyn}$ extracts multi-level features, which are upsampled by $\mathbf{U}_{dy}$ (DySample). Slim-Neck $\mathbf{F}_{slim}$ aggregates features across top-down and bottom-up paths, and the detection head $\mathbf{H}_{det}$ produces multi-scale predictions $\mathbf{Y}$. The pipeline is:

$$\mathbf{Y} = \mathbf{H}_{det}(\mathbf{F}_{slim}(\mathbf{U}_{dy}(\mathbf{B}_{dyn}(\mathbf{X})))) \,. \tag{1}$$

3.2 D-HGNet: Dynamic High-Performance Backbone

As the backbone of SlimDy-YOLO, D-HGNet consists of three components: HGStem [3], stacked Dynamic_HGBlocks [24], and DWConv-based downsampling [28]. HGStem captures shallow fracture-relevant structural cues such as local edges, fine textures, and subtle discontinuities. Dynamic_HGBlocks then extract deeper features by adapting to the shallow features, helping the backbone represent radiograph-specific contextual patterns around fracture-like discontinuities and growth-plate-like structures. Between stages, DWConv downsamples features through grouped convolutions, enabling the backbone to build multi-level representations with minimal computational cost.

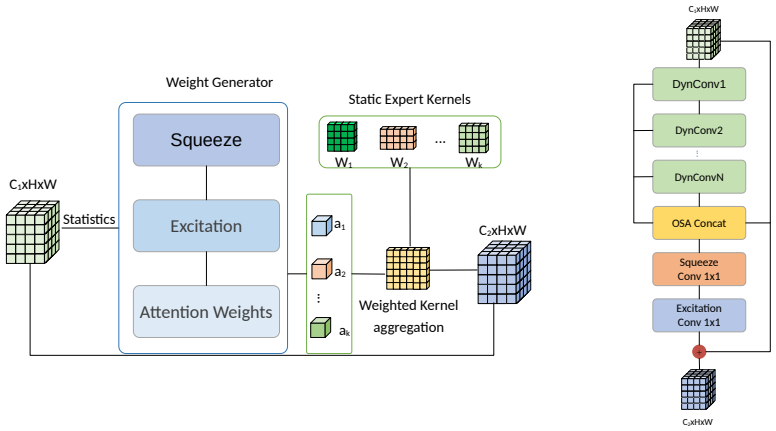
HGStem HGStem uses a dual-branch design to initialize shallow features. Unlike conventional large-stride stems that lose local information, this design preserves structural details relevant to fracture patterns. The input image $\mathbf{X}$ passes through an initial stem layer to produce an intermediate feature map $\mathbf{F}_1$. Features from a pooling branch and a small-kernel convolution branch are then concatenated and fused [21]. The feature initialization process can be written as follows:

$$\mathbf{F}_1 = \mathrm{Stem}_1(\mathbf{X}), \tag{2}$$

$$\mathbf{F}_{stem} = \mathrm{Stem}_4\Big(\mathrm{Stem}_3\big(\,\mathrm{Concat}[\mathrm{MaxPool}(\mathbf{F}_1),\, \mathrm{Stem}_{2b}(\mathrm{Stem}_{2a}(\mathbf{F}_1))]\big)\Big). \tag{3}$$

Dynamic_HGBlock Built upon the shallow features generated by HGStem, Dynamic_HGBlocks perform deeper feature extraction in an input-adaptive manner. As illustrated in Fig. 2, the core operation is dynamic convolution, which replaces fixed kernels with an input-conditioned aggregation of K expert kernels [24]. For an input feature $\mathbf{x}$, the effective kernel is determined by dynamic coefficients $\alpha_k(\mathbf{x})$, predicted from the input and normalized by Softmax:

$$\mathbf{y} = \left(\sum_{k=1}^{K} \alpha_k(\mathbf{x}) \cdot \mathbf{W}_k\right) * \mathbf{x}, \quad \text{s.t.} \ \sum_{k=1}^{K} \alpha_k(\mathbf{x}) = 1. \tag{4}$$



(a) Dynamic convolution (b) Dynamic_HGBlock

Fig. 2: Overview of Dynamic_HGBlock and dynamic convolution.

This mechanism enables the feature extractor to adapt its responses to local X-ray image content and produces contextual features that are more sensitive to subtle discontinuities and nearby anatomical structures. To facilitate gradient propagation and feature reuse, Dynamic_HGBlock uses a cascaded generation strategy, where features are progressively refined through n dynamic convolution layers. The input and all intermediate outputs are then concatenated, followed by squeeze-and-excitation channel recalibration and a residual shortcut (Fig. 2b):

$$\mathbf{F}_{out} = \mathrm{Conv}^{\mathrm{ex}}_{1\times1}\big(\mathrm{Conv}^{\mathrm{sq}}_{1\times1}([\mathbf{X};\mathbf{Y}_1;\ldots;\mathbf{Y}_n])\big) + \mathrm{P}(\mathbf{X}), \tag{5}$$

where $\mathrm{Conv}^{\mathrm{sq}}_{1\times1}$ squeezes the channel dimension to $C_2/2$, and $\mathrm{Conv}^{\mathrm{ex}}_{1\times1}$ excites it back to C_2. $\mathrm{P}(\cdot)$ denotes the shortcut projection (identity if $C_1 = C_2$, otherwise a 1×1 projection) for residual addition.

3.3 Slim-Neck: Lightweight Feature Fusion Module

D-HGNet extracts multi-level features with different characteristics: shallow features capture local structural details relevant to fractures, while deeper features encode broader context. To fuse these features efficiently for real-time clinical use, we use Slim-Neck [10] as the neck of SlimDy-YOLO. Rather than applying dense convolution across the fusion path, Slim-Neck applies lightweight modules for top-down and bottom-up aggregation. This design lets shallow features benefit from deeper context during cross-scale fusion without adding significant computation.

Within Slim-Neck, GSConv reduces the redundancy of dense convolution when fusing D-HGNet's multi-level features. Instead of applying full spatial-channel transformation to all channels, GSConv decomposes feature mixing into

a lightweight process: standard convolution first performs channel interaction, depthwise convolution then injects spatial information at lower cost, and channel shuffle further restores cross-channel communication after concatenation. Given an input feature map $\mathbf{X}$, its computation can be written as

$$\mathbf{F}_{sc} = \text{ConvBlock}_{1\times1}^{C_{out}/2}(\mathbf{X}), \tag{6}$$

$$\mathbf{F}_{dsc} = \text{DWConv}_{5\times5}^{C_{out}/2}(\mathbf{F}_{sc}), \tag{7}$$

$$\mathbf{Y} = \text{Shuffle}([\mathbf{F}_{sc}; \mathbf{F}_{dsc}]). \tag{8}$$

By decomposing dense feature mixing into lightweight channel interaction, spatial refinement, and channel rearrangement, GSConv offers an efficient approach to fuse D-HGNet's multi-level features during cross-scale aggregation. To improve feature propagation without adding heavy computation, VoVGSCSP follows GSConv-based transformation. Given an input feature map $\mathbf{X}$, the feature channels are split into two branches: one branch passes through stacked GSBottleneck operations, while the other serves as a lightweight shortcut that maintains direct feature flow:

$$\mathbf{F}_{deep} = \text{H}_{GSB}(\text{Conv}_1(\mathbf{X})), \tag{9}$$

$$\mathbf{F}_{proj} = \text{Conv}_2(\mathbf{X}), \tag{10}$$

$$\mathbf{Y} = \text{Conv}_{fusion}([\mathbf{F}_{deep}; \mathbf{F}_{proj}]), \tag{11}$$

where $\mathcal{H}_{GSB}(\cdot)$ denotes the transformation of stacked GSBottleneck units. This partial-channel aggregation strategy keeps direct feature flow in one branch while refining the other. This design suits D-HGNet's multi-level outputs: shallow structural cues remain intact while deeper context gets integrated. The fused feature maps then move to the upsampling path, where DySample reconstructs features during top-down alignment.

3.4 DySample: Content-Aware Upsampling Module

During top-down fusion in Slim-Neck, deep features must be upsampled and aligned with higher-resolution feature maps. Fixed interpolation is efficient, but it may smooth subtle local structures that are useful for fracture localization. To preserve these cues without increasing the cost of the upsampling path, we adopt DySample as a content-aware upsampling module [12, 6]. Given an input feature map $\mathbf{X}$, the upsampling process is expressed as

$$\mathbf{O} = \text{Linear}(\mathbf{X}), \qquad \mathbf{G} = \mathbf{G}_{init} + 0.25\,\mathbf{O}, \tag{12}$$

$$\mathbf{G}_{high} = \text{PixelShuffle}(\mathbf{G}, s), \qquad \mathbf{Y} = \text{GridSample}(\mathbf{X}_{grp}, \mathbf{G}_{high}). \tag{13}$$

To reduce the computational cost, the channel dimension is divided into g groups and folded into the batch dimension, giving $\mathbf{X}_{grp} \in \mathbb{R}^{(B\cdot g)\times(C/g)\times H\times W}$. This formulation allows group-wise `grid_sample` to be executed as a single batched sampling operation. In implementation, we set `align_corners=False` to avoid coordinate drift and use `padding_mode=border` to reduce boundary artifacts by replicating border values.

4 Results and Discussion

4.1 Dataset and Data Augmentation

This study uses the GRAZPEDWRI-DX dataset, a public benchmark for pediatric wrist trauma X-ray analysis [14]. It contains 20,327 annotated radiographs from 6,091 patients collected at the University Hospital Graz between 2008 and 2018, with 67,771 labeled objects across nine categories: *fracture*, *bone lesion*, *bone anomaly*, *foreign body*, *metal*, *periosteal reaction*, *pronator sign*, *soft tissue*, and *text*. The multi-class annotations allow evaluation of fracture localization in the presence of visually related anatomical findings and image artifacts. To avoid patient-level leakage, train/validation/test splits were created at the patient level with a 7:2:1 ratio, and all radiographs from the same patient were assigned to only one subset. The final training, validation, and test sets contained 14,228, 4,065, and 2,034 images, respectively.

To improve robustness to radiographic appearance variation, we use horizontal flipping and bounded intensity/noise augmentations. Horizontal flipping doubles the training set with consistent bounding-box coordinate updates. We then augment 60% of the combined training set using clinically plausible brightness, contrast, gamma, CLAHE, Gaussian noise, salt-and-pepper noise, and blur transformations. Large rotations, elastic deformation, and perspective distortion are not used to avoid altering wrist anatomy or obscuring fracture lines.

4.2 Evaluation Metrics

We use mAP_{50} as the primary detection metric, computed at an IoU threshold of 0.5. Precision, Recall, and F1-score are reported as supporting measures of detection behavior, where F1 summarizes the balance between false positives and false negatives. Parameters, GFLOPs, and FPS quantify model size, theoretical computation, and measured throughput, respectively. Together, these metrics evaluate whether a detector improves fracture localization while satisfying the real-time screening constraint.

4.3 Baseline Selection and Experimental Setup

All experiments were conducted on a workstation running Windows Server 2019, equipped with an Intel(R) Xeon(R) Gold 6252 CPU and an NVIDIA GeForce RTX 4090 GPU. The software environment used Python and PyCharm with

PyTorch 2.2.2 accelerated by CUDA 11.8. Before evaluating the proposed modules, we compared YOLOv11 models across five scales under the same dataset setting. As shown in Table 1, YOLOv11m achieves the highest mAP_{50} among the tested scales, while larger models do not provide a clear accuracy gain relative to their additional cost. We therefore use YOLOv11m as the base detector for the subsequent module-level analysis.

Table 1: Performance comparison of YOLOv11 models across different scales.

Model	P	R	F1	mAP_{50}	GFLOPs	Params (M)	FPS
YOLOv11n	0.7807	0.5713	0.6598	0.6330	6.3	2.58	380.50
YOLOv11s	0.7653	0.5480	0.6387	0.6246	21.3	9.42	342.91
YOLOv11m	0.7313	**0.6296**	**0.6767**	**0.6511**	67.7	20.04	313.89
YOLOv11l	0.7381	0.6008	0.6625	0.6460	86.6	25.29	290.08
YOLOv11x	0.6748	0.5971	0.6335	0.6365	194.5	56.84	202.55

Building on YOLOv11m, we evaluated initialization strategy, optimizer, and input resolution before the module ablations. Training from scratch slightly outperformed COCO pre-training in mAP_{50} and FPS, likely due to the domain gap between pediatric wrist radiographs and natural images. SGD outperformed Adam, and increasing the resolution to 1024×1024 produced only a marginal mAP_{50} gain while reducing FPS substantially. We therefore used training from scratch with SGD at 640×640 resolution for all subsequent experiments, using 100 epochs, batch size 32, initial learning rate 0.01, final learning-rate factor 0.01, and early-stopping patience 10.

4.4 Ablation Studies

Stepwise Effectiveness Analysis The stepwise ablation in Table 2 tests whether the proposed modules support the two design goals introduced earlier. To reduce the influence of training randomness, each setting is repeated with three random seeds, and the averaged results are reported. Adding D-HGNet evaluates adaptive fracture-evidence extraction: compared with the YOLOv11m baseline, mAP_{50} increases from 0.6482 to 0.6656, while GFLOPs decrease from 67.7 to 47.0 and parameters decrease from 20.04M to 17.82M. This result supports the use of an input-adaptive backbone for radiograph-specific feature extraction, without attributing the gain to a single clinical mechanism. Adding Slim-Neck evaluates efficient multi-scale evidence preservation through lightweight fusion, further increasing mAP_{50} to 0.6714 and reducing GFLOPs to 40.5. Finally, adding DySample improves the final model to 0.6843 mAP_{50} and 0.6943 F1. Although FPS decreases to 267.98, the throughput remains practical under the tested inference setting.

Comparative Analysis of Neck Architectures The neck comparison in Table 3 examines how different fusion designs affect efficient multi-scale evidence preservation when the backbone is fixed as D-HGNet. Heavier fusion modules do

Table 2: Stepwise ablation study on the proposed improvements.

Algorithm	F1	mAP_{50}	GFLOPs	Params (M)	FPS
Baseline (YOLOv11m)	0.6783	0.6482	67.7	20.04	313.56
+ D-HGNet	0.6629	0.6656	47.0	17.82	304.81
+ Slim-Neck	0.6632	0.6714	**40.5**	**16.80**	290.13
+ DySample	**0.6943**	**0.6843**	**40.5**	16.83	267.98

not consistently improve this task: CSFCN [11], MAFPN [27], and GFPN [22] increase computation or parameters without improving mAP_{50} over Slim-Neck, and ASF-Neck [20] provides only a marginal gain over several alternatives at 56.8 GFLOPs. At the other extreme, HSFPN reduces computation to 34.2 GFLOPs but drops mAP_{50} to 65.86%, suggesting that aggressive compression can discard useful multi-scale cues. Among the tested designs, Slim-Neck provides the best mAP_{50} of 67.34% with 40.5 GFLOPs and 16.80M parameters, supporting its role as the lightweight fusion component.

Table 3: Performance comparison of different Neck architectures based on D-HGNet backbone.

Neck Type	F1	mAP_{50}	GFLOPs	Params (M)	FPS
ASF-Neck	0.6640	0.6716	56.8	19.08	262.43
BiFPN	0.6652	0.6680	48.3	13.58	282.65
CSFCN	0.6649	0.6656	56.4	20.16	260.50
GFPN	0.6596	0.6630	73.0	30.35	282.31
HSFPN	0.6555	0.6586	34.2	11.83	296.83
MAFPN	0.6644	0.6638	63.2	25.29	263.88
Slim-Neck (Ours)	0.6648	**0.6734**	40.5	16.80	290.26

Evaluation of Upsampling Operators The upsampling comparison in Table 4 evaluates whether content-aware alignment helps preserve high-resolution fracture-related cues during top-down fusion. CARAFE [19] reaches 67.99% mAP_{50}, but its throughput drops to 7.96 FPS, making it unsuitable for the real-time constraint in our setting. EUCB increases computation to 44.7 GFLOPs with lower mAP_{50}, while Converse2d keeps GFLOPs at 40.5 but reduces throughput to 53.19 FPS. DySample [12] achieves the highest mAP_{50} of 68.47% with 40.5 GFLOPs and 265.96 FPS, providing the best tested accuracy-cost balance for the upsampling stage.

4.5 Comparative experiments

Table 5 compares SlimDy-YOLO with mainstream detector baselines under the same experimental setting. Relative to the YOLOv11m base detector, SlimDy-YOLO improves mAP_{50} from 0.6511 to 0.6847, reduces GFLOPs from 67.7 to 40.5, and reduces parameters from 20.04M to 16.83M. This result supports the coordinated redesign of the backbone, neck, and upsampling path rather

Table 4: Comparative analysis of upsampling operators based on the D-HGNet and Slim-Neck.

Upsampling	F1	mAP_{50}	GFLOPs	Params (M)	FPS
CARAFE	0.7021	0.6799	40.8	16.98	7.96
Converse2d	0.6640	0.6716	40.5	16.81	53.19
EUCB	0.6596	0.6689	44.7	17.33	226.08
DySample (Ours)	0.6918	**0.6847**	**40.5**	16.83	265.96

Table 5: Performance comparison with mainstream detection baselines.

Model	P	R	F1	mAP_{50}	GFLOPs	Params (M)	FPS
YOLOv26m	0.7496	0.5661	0.6449	0.6244	67.9	20.36	286.51
YOLOv13s	0.7385	0.5909	0.6565	0.6209	20.1	9.00	342.82
YOLOv12m	0.6789	0.6115	0.6435	0.6262	59.5	19.58	199.79
YOLOv10m	0.6743	0.5952	0.6324	0.6281	58.9	15.32	310.89
YOLOv9m	0.7132	0.6116	0.6584	0.6384	76.5	20.02	229.79
YOLOv8m	0.7293	0.5859	0.6498	0.6338	78.7	25.84	294.15
YOLOv5m	0.6950	0.6006	0.6444	0.6287	64.0	25.05	340.25
YOLOv11m	0.7313	0.6296	0.6767	0.6511	67.7	20.04	313.89
RT-DETR-R50	0.6683	0.6072	0.6362	0.6298	125.7	41.95	161.05
SlimDy-YOLO	0.7068	**0.6776**	**0.6918**	**0.6847**	40.50	16.83	265.96

than simply scaling the detector. RT-DETR-R50 has substantially higher computation and parameter count in this setting, while its mAP_{50} remains below SlimDy-YOLO. Overall, the comparison indicates that SlimDy-YOLO improves the accuracy-cost balance for pediatric wrist fracture detection under the tested hardware and implementation setting.

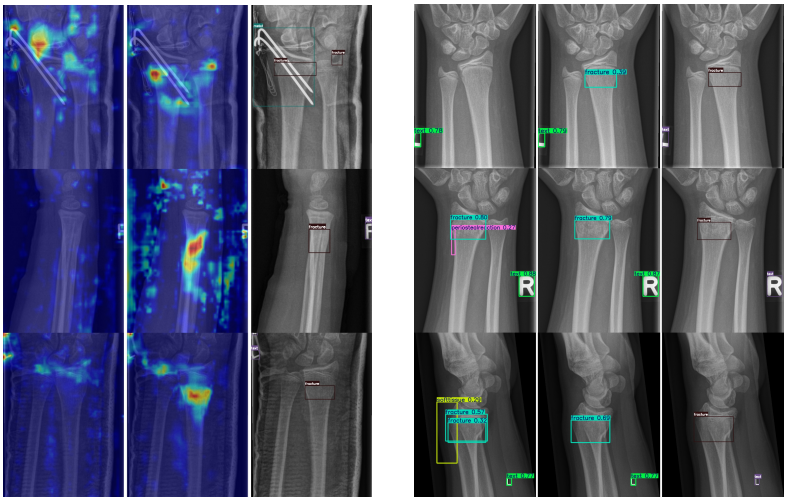
Table 6 further summarizes reported results from recent pediatric wrist fracture detection studies. Because these studies may differ in implementation details and evaluation protocols, the table is used as contextual evidence rather than as a perfectly controlled head-to-head comparison.

Table 6: Comparison of prior studies.

Studies	Methods Used	mAP_{50}	GFLOPs	Params (M)
Chien et al. [4]	YOLOv8 + ResCBAM	0.6578	196.2	53.87
Ju et al. [8]	YOLOv8 + SE-M3	0.6707	165.6	43.85
Ferdi et al. [5]	G-YOLOv11	0.5350	21.0	7.79
Qiu et al. [16]	FracDet-v11	0.6480	18.10	7.90
Tariq et al. [18]	YOLOv11 + SE_BLOCK	0.6510	172.3	49.32
Ours	SlimDy-YOLO	**0.6847**	40.50	16.83

4.6 Model Evaluation and Analysis

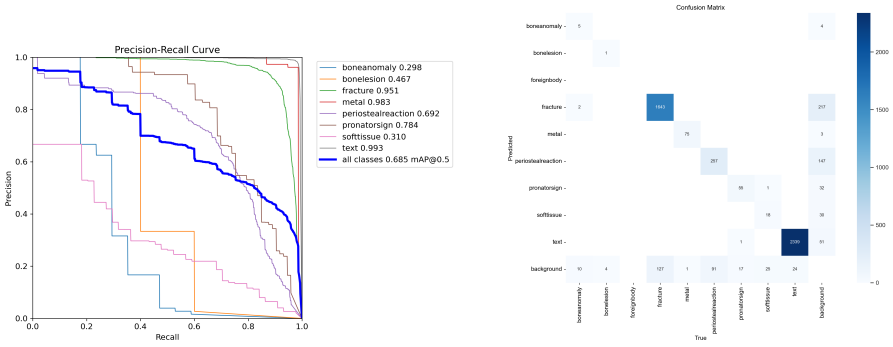
Figure 4 summarizes class-wise PR curves and the confusion matrix. The results show that categories with clearer visual patterns, such as *text*, *metal*, and *fracture*, achieve stronger precision-recall behavior, whereas subtle categories such as *boneanomaly* and *softtissue* remain more difficult. The confusion matrix indicates that many remaining errors are related to background confusion, including



(a) Grad-CAM comparison. (b) Detection results.

Fig. 3: Qualitative visualization for attention localization and detection robustness (L–R: Baseline, Improved, GT).

missed subtle findings and false detections on ambiguous structures. The qualitative examples in Fig. 3 are consistent with this pattern: compared with the baseline, SlimDy-YOLO shows more localized activation around lesion regions and produces fewer missed or spurious boxes in the shown cases. These visual results support the weak-evidence localization narrative, while the quantitative tables remain the primary evidence for model effectiveness.



(a) Per-class PR curves at IoU=0.5 (AP@0.5). (b) Confusion matrix (rows: predicted, columns: ground truth).

Fig. 4: Class-wise performance and error patterns.

5 Conclusions

This paper presents SlimDy-YOLO for pediatric wrist fracture detection, framing the task as weak radiographic evidence detection under real-time screen-

ing constraints. The method addresses two core problems: adaptive fracture-evidence extraction through D-HGNet and efficient multi-scale evidence preservation through Slim-Neck and DySample. On GRAZPEDWRI-DX, SlimDy-YOLO achieves 68.47% mAP_{50} with 40.5 GFLOPs and 16.83M parameters, improving over the YOLOv11m baseline while reducing computation and model size. Ablation, neck, upsampling, and baseline comparisons support the coordinated backbone-neck-upsampling design, and qualitative results provide visual support for improved weak-evidence localization. Future work will focus on external validation across institutions and further analysis of difficult false-positive and false-negative cases.

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